

Practical 2

Question 1

My Workspace > Practical2.sql

```
1 -----
2 ---1. INNER JOIN: Orders with Customer and Product Names
3 --Question:
4 --List all orders along with the customer name and product name.
5 --Expected Output Columns:OrderID, OrderDate, CustomerName, ProductName, Quantity
6 SELECT B.ORDERID,
7        B.ORDERDATE,
8        A.CUSTOMERNAME,
9        C.PRODUCTNAME,
10       B.QUANTITY
11 FROM
12 PRACTICAL2.DATASET.CUSTOMERS AS A
13 INNER JOIN PRACTICAL2.DATASET.ORDERS AS B
14 ON A.CUSTOMERID=B.CUSTOMERID
15 INNER JOIN PRACTICAL2.DATASET.PRODUCTS AS C
16 ON C.PRODUCTID=B.PRODUCTID;
```

Results (3 minutes ago)

Table Chart 4,000 rows 844ms

	# ORDERID	ORDERDATE	A CUSTOMERNAME	A PRODUCTNAME	# QUANTITY
1	1	2023-06-10	Customer_1251	Product_2014	10
2	2	2023-12-07	Customer_1236	Product_2004	5
3	3	2024-10-26	Customer_1170	Product_2171	9
4	4	2023-02-17	Customer_1344	Product_2007	2

Question 2

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18 ---2. INNER JOIN: Customers Who Placed Orders
19 --Question: Which customers have placed at least one order?
20 --Expected Output Columns:
21 --CustomerID, CustomerName, Country, OrderID, OrderDate
22
23 SELECT A.CUSTOMERID,
24        A.CUSTOMERNAME,
25        A.COUNTRY,
26        B.ORDERID,
27        B.ORDERDATE
28 FROM PRACTICAL2.DATASET.CUSTOMERS AS A
29 INNER JOIN PRACTICAL2.DATASET.ORDERS AS B
30 ON A.CUSTOMERID=B.CUSTOMERID;
```

Results (just now)

Table Chart 4,000 rows 32ms

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	ORDERDATE
1	1251	Customer_1251	Germany	1	2023-06-10
2	1236	Customer_1236	Australia	2	2023-12-07
3	1170	Customer_1170	Germany	3	2024-10-26
4	1344	Customer_1344	Canada	4	2023-02-17
5	1319	Customer_1319	USA	5	2024-11-06

Question 3

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33  --3. LEFT JOIN: ALL Customers and Their Orders
34  --Question: List all customers and any orders they might have
    placed. Include customers who -have not placed any orders.
35  --Expected Output Columns:CustomerID, CustomerName, Country,
    OrderID, OrderDate, ProductID, Quantity
36
37  SELECT A.CUSTOMERID,
38         A.CUSTOMERNAME,
39         A.COUNTRY,
40         B.ORDERID,
41         B.ORDERDATE,
42         B.PRODUCTID,
43         B.QUANTITY
44
45  FROM PRACTICAL2.DATASET.CUSTOMERS AS A
46  LEFT JOIN PRACTICAL2.DATASET.ORDERS AS B
47  ON A.CUSTOMERID=B.CUSTOMERID;

```

Results (1 minute ago)

Table Chart 4,000 rows 575ms

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	ORE
1	1251	Customer_1251	Germany	1	2023-(
2	1236	Customer_1236	Australia	2	2023-(
3	1170	Customer_1170	Germany	3	2024-(

Question 4

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49  --4. LEFT JOIN: Product Order Count
50  --Question: List all products and how many times each was ordered
    (if any).
51  --Expected Output Columns:ProductID, ProductName, TotalOrders
    (TotalOrders is the count of how many times the product appears
    in orders)
52
53  SELECT B.PRODUCTID,
54         C.PRODUCTNAME,
55         COUNT(B.ORDERID) AS TotalOrders
56  FROM PRACTICAL2.DATASET.ORDERS AS B
57  LEFT JOIN PRACTICAL2.DATASET.PRODUCTS AS C
58  ON B.PRODUCTID=C.PRODUCTID
59  GROUP BY B.PRODUCTID,C.PRODUCTNAME;|

```

Results (1 minute ago)

Table Chart 200 rows 30ms

	# PRODUCTID	A PRODUCTNAME	# TOTALORDERS
1	2105	Product_2105	14
2	2063	Product_2063	24
3	2059	Product_2059	17

Question 5

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```

61  ---5. RIGHT JOIN: Orders with Product Info (Include Products Not Ordered)
62  --Question: Find all orders along with product details, including any products
    that might not have been ordered.Expected Output Columns:OrderID, OrderDate,
    ProductID, ProductName, Price, Quantity
63  SELECT B.ORDERID,
64         B.ORDERDATE,
65         C.PRODUCTID,
66         C.PRODUCTNAME,
67         C.PRICE,
68         B.QUANTITY
69  FROM PRACTICAL2.DATASET.ORDERS AS B
70  RIGHT JOIN PRACTICAL2.DATASET.PRODUCTS AS C
71  ON B.PRODUCTID=C.PRODUCTID;
72

```

Results (just now)

Table Chart 4,000 rows 23ms

	# ORDERID	ORDERDATE	# PRODUCTID	PRODUCTNAME	# PRICE	# QUANTITY
1	1	2023-06-10	2014	Product_2014	522	10
2	2	2023-12-07	2004	Product_2004	1996	5
3	3	2024-10-26	2171	Product_2171	76	9
4	4	2023-02-17	2007	Product_2007	156	2

Question 6

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```

72
73  ---6. RIGHT JOIN: Customer Info with Orders (Include All Customers)
74  --Question: Which customers have made orders, and include customers even if they have never
    placed an order.Expected Output Columns:CustomerID, CustomerName, Country, OrderID, OrderDate,
    ProductID, Quantity
75
76  SELECT B.CUSTOMERID,
77         A.CUSTOMERNAME,
78         A.COUNTRY,
79         B.ORDERID,
80         B.ORDERDATE,
81         B.PRODUCTID,
82         B.QUANTITY
83  FROM PRACTICAL2.DATASET.CUSTOMERS AS A
84  RIGHT JOIN PRACTICAL2.DATASET.ORDERS AS B
85  ON A.CUSTOMERID=B.CUSTOMERID;
86

```

Results (1 minute ago)

Table Chart 4,000 rows 573ms

	# CUSTOMERID	CUSTOMERNAME	COUNTRY	# ORDERID	ORDERDATE	# PRODUCTID	# QUA
1	1251	Customer_1251	Germany	1	2023-06-10	2014	
2	1236	Customer_1236	Australia	2	2023-12-07	2004	
3	1170	Customer_1170	Germany	3	2024-10-26	2171	

Question 7

87 --7. FULL OUTER JOIN: All Customers and All Orders
88 --Question: List all customers and orders, showing NULLs where customers have not ordered or
where orders have no customer info.
89 --Expected Output Columns: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID,
Quantity
90
91 SELECT A.CUSTOMERID,
92 A.CUSTOMERNAME,
93 A.COUNTRY,
94 B.ORDERID,
95 B.ORDERDATE,
96 B.PRODUCTID,
97 B.QUANTITY
98 FROM PRACTICAL2.DATASET.CUSTOMERS AS A
99 FULL OUTER JOIN PRACTICAL2.DATASET.ORDERS AS B
100 ON A.CUSTOMERID=B.CUSTOMERID;

Results (1 minute ago)

Table Chart 4,000 rows 33ms

	CUSTOMERID	CUSTOMERNAME	COUNTRY	ORDERID	ORDERDATE	PRODUCTID	QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	1
2	1236	Customer_1236	Australia	2	2023-12-07	2004	
3	1170	Customer_1170	Germany	3	2024-10-26	2171	

Question 8

102 --8. FULL OUTER JOIN: All Products and Orders
103 --Question: List all products and orders, showing NULLs where products were never ordered or orders are missing product
info. Expected Output Columns: ProductID, ProductName, Price, OrderID, OrderDate, CustomerID, Quantity
104
105 SELECT C.PRODUCTID,
106 C.PRODUCTNAME,
107 C.PRICE,
108 B.ORDERID,
109 B.ORDERDATE,
110 B.CUSTOMERID,
111 B.QUANTITY
112 FROM PRACTICAL2.DATASET.ORDERS AS B
113 FULL OUTER JOIN PRACTICAL2.DATASET.PRODUCTS AS C
114 ON B.PRODUCTID = C.PRODUCTID;
115
116 | Ctrl+I to generate
117

Results (just now)

Table Chart 4,000 rows 22ms

	PRODUCTID	PRODUCTNAME	PRICE	ORDERID	ORDERDATE	CUSTOMERID	QUANTITY
1	2014	Product_2014	522	1	2023-06-10	1251	10
2	2004	Product_2004	1996	2	2023-12-07	1236	5
3	2171	Product_2171	76	3	2024-10-26	1170	9